

Animatronic User Guide

A thing made by Andrew Langley (andrew@langleycreations.com)

Overview

This system brings a restored animatronic character to life. It runs on an embedded Linux computer and lets the character hold spoken conversations, perform pre-programmed shows, and live puppeteering, all of which is controllable from a web interface.

Safety

- If the animatronic is pneumatic, never raise the regulator above the PSI specified for your particular character. For example, Kermit should run at **60 PSI**.
- Only power the system with the provided power supply. For example, Kermit requires **24v 120w** for the electronics.
- If you need to open the electronics case or unplug any cables (other than audio or USB), disconnect power completely to avoid any permanent damage.

Physical configuration

Connection	Required?	Notes
Power adapter	Yes	Powers the controller. The system starts when power is connected after about 30 to 60 seconds of boot time.
Air compressor	Yes, if the animatronic is pneumatic.	Set the regulator to the correct PSI. Higher pressure may cause problems.
3.5 mm audio jack	Yes, for sound	Line-level output. It needs a powered speaker or an amplifier, not headphones .

4-mic array with LEDs (ReSpeaker XVF3800), USB-A	Yes, if the animatronic supports voice control	Used for speech recognition. Its listening direction is tuned to where the audience stands, so don't reposition it inside the figure.
USB thumb drive, USB-A	Recommended	Holds config.cfg and the shows folder. See Software configuration.
Wireless gamepad dongle, USB-A	Optional	For live puppeteering. Plug-and-play.
USB keyboard, USB-A	Optional	For live puppeteering with keyboard keys.
Program Blue cable (USB-serial), USB-A	Optional	For animating in Program Blue on a PC. Runs through a null-modem adapter with two USB to RS232 adapters .
USB-C port	Optional	Connects to a computer or MIDI host. The character appears as a MIDI device named "Animatronic MIDI". Note that ONLY the USB-C port on the character will provide a MIDI connection.

Booting

The system starts automatically when the power adapter is connected. Allow about **30 to 60 seconds** before it responds to voice.

Talking to the character

Say the character's **wake word**, wait for the light ring on the microphone array to turn on, then speak your question or command. For example, *"Okay Kermit"*.

Status LED light (located on the black round USB microphone)

Light	Meaning
Off	Idle and waiting for the wake word, or speaking
Solid	Listening to you
Slow breathing	Thinking: waiting for the AI response

Connecting to WiFi

Once the character has joined a network with a password, it remembers it and can rejoin it later without the password.

Option 1: config.cfg on the USB drive. Edit the [WiFi] section, then plug the drive back in. It is picked up immediately, with no restart needed:

Option 2: the web interface. Click the WiFi icon in the upper-left corner, select a network, enter the password if needed, and click **Connect**.

Option 3: voice command. Say the wake word, then "Connect to WiFi <network name>". The character picks the visible network whose name sounds closest. Passwords can't be spoken, so this only works for networks it has joined before. For a new network, use Option 1 or 2.

If a connection fails, the character says whether the password was wrong, the network wasn't found, or the connection failed for another reason.

Accessing the web interface

Any phone or computer on the same network as the character can control it from a web browser.

1. Ask for its address: say the wake word, then "What's your IP address?" The character reads it out one digit at a time.
2. Type it into a browser, for example <http://192.168.0.5>

You may also be able to reach it by name. For example: <http://kermit.local> or <http://pasqually.local> This may or may not work depending on how the network is configured.

Software configuration

Settings live in **config.cfg** on the USB drive, and contain the values you'll change often: WiFi, API keys, the AI persona, and the voice parameters.

There are two ways to change config.cfg:

- **On the USB drive.** Edit **config.cfg** at the top level of the drive on a computer, then plug the drive into the character. Settings take effect as soon as the drive is detected.
- **In the web interface.** Click the gear icon in the upper right, change values, and click **Save**. Changes apply immediately and are written to both the USB drive and the copy on the device.

ElevenLabs voice

ElevenLabs provides the character's main voice. Without it, the character uses the offline voice (see Offline mode).

Setting ([TextToSpeech])	Default	What it does
ElevenLabsKey	None	API key from your ElevenLabs account . Required.
ElevenLabsVoiceID	None	ID of the voice to use . Required.
ElevenLabsStability	0.35	0 to 1. Lower is more expressive and varied; higher is more consistent and flatter.
ElevenLabsSimilarityBoost	0.75	0 to 1. How closely to stick to the original voice. Too high can reproduce noise from

		the source recordings. Keep the spelling exactly as shown.
ElevenLabsStyle	0.3	0 to 1. Exaggerates the voice's style. Higher values add personality but can reduce stability and add delay.
ElevenLabsUseHighQualitySlowModel	0	1 uses ElevenLabs' higher-quality model, which adds noticeable delay before the character speaks. 0 uses the fast model.

NOTE: If ElevenLabs has not been configured or cannot connect, the system falls back to a locally cloned voice for the character instead. The quality of this voice won't sound as natural as ElevenLabs, but it's meant to work as a backup so that the system doesn't break.

AI services

The character can use Anthropic Claude, OpenAI ChatGPT, or DeepSeek. Enter an API key for at least one if you want to use online AI services. If more than one has a key, they're tried in this order and the first that answers is used: **Claude, then ChatGPT, then DeepSeek.**

You do not need to enter values for all of them. Just for the provider you want to use.

Section	Setting	Notes
[Anthropic]	AnthropicKey	API key
[Anthropic]	AnthropicModel	Model name. Blank uses <i>claude-sonnet-4-6</i> .
[ChatGPT]	OpenAIKey	API key

[ChatGPT]	OpenAIModel	Model name. Blank uses <i>gpt-4o-mini</i> .
[DeepSeek]	DeepSeekAPIKey	API key
[DeepSeek]	DeepSeekModel	Model name, default <i>deepseek-chat</i>

The [LLM] section holds two character descriptions:

- **LLMContextShort** says *who* the character is: identity, friends, history and "answer in one or two short sentences". It is the only description the offline model sees, so keep it short and put the essentials here. For example: "You are Kermit the Frog. Speak in first person as yourself..."
- **LLMContext** says *how* the character behaves: personality, catchphrases, topics to avoid. The online services receive both descriptions together.

Other settings

Setting	Default	What it does
[Wakeword] Threshold	0.3	How sure the character must be that it heard its wake word, from 0 to 1. Lower responds more easily but triggers falsely more often. Raise it in noisy rooms.
[LED] LEDDisable	0	1 keeps the microphone light ring dark at all times.

Both can be set on the USB drive or in the web configuration editor.

NOTE: If none of the AI services have been configured, or if all of them fail to connect, the system will fall back to an AI model running locally on the linux computer. The results won't be as high quality as the online models, but will allow the system to continue to function as a backup solution so that the character will still respond.

Manual movement control

You can puppeteer the character live from the web keypad, a wireless gamepad or a USB keyboard. Each movement stays on for as long as you hold its button or key.

Mirrored mode swaps left and right, so a routine can be performed facing the other way. Turn it on with the checkbox in the web interface.

Playing shows

Shows play from the USB drive's **shows** folder, or from the device's backup copy when no drive is attached.

- **Web interface:** tap the show selector, pick a show, then press play.
- **Voice:** say the wake word, then "Play <show name>" or "Sing <show name>". "Sing a song" picks one at random.

Animating custom shows

A show is an audio track plus a record of which movements are on at each moment. The system accepts two formats:

- **Program Blue show** (.shw): a single file containing both the audio and the movements.
- **MIDI + audio pair:** a .mid file for the movements and a matching audio file (.mp3, .wav or .ogg).

Program Blue

1. Connect the Program Blue cable from the USB-A port on the back of the character to your PC's serial port through the provided null-modem adapter. Additional hardware like the BlueSpider box is not required.
2. Start Program Blue and open the character's template from the USB drive's "shows" directory. Keep the frame rate at **40 frames per second**; other rates will play back at the wrong speed.

When Program Blue connects, the character responds to it and moves live as you play or scrub the show.

MIDI

1. Connect a USB cable from the **USB-C** port on the back of the character to your computer or MIDI hardware. It appears as a MIDI device named **Animatronic MIDI**.
2. Each movement is one note.

3. **To record live:** arm a MIDI track in your sequencer and press record, then perform with the web keypad, gamepad or keyboard. Every movement you make is sent out as a note.
4. Export the result as a standard MIDI file (.mid) and line it up with your audio from time zero.

For saved .mid files, use a note velocity of **90 or higher**. Notes below 90 are treated as "off" when a show file is played back, even though they work when sent live.

Adding custom shows

Copy the show files into the **shows** folder at the top level of the USB drive, then plug the drive into the character. New shows appear in the web interface's show list and can be requested by voice straight away.

Format	Files needed	Example
Program Blue	One .shw file	Rainbow Connection.shw
MIDI + audio	A .mid file and an .mp3, .wav or .ogg file with exactly the same name	Rainbow Connection.mid + Rainbow Connection.mp3

File-naming tips:

- Use the lowercase .mid extension. .midi and .MID files are not picked up reliably.
- The file name is what people say to request the show by voice, so use the song's real title.
- A MIDI file with no matching audio file (or vice versa) won't appear in the list.

Web upload: From the show selector page on the web interface, you have the option to upload files directly.

Offline mode

As mentioned previously, the character keeps talking with no internet connection or no API keys, using an AI model and a backup voice that both run on the device. Answers are simpler and shorter than online, but it stays in character and continues to function.

Troubleshooting

Overall, when a problem happens, just try rebooting! Power the system off and back on and see if that solves the problem.

Problem	Things to check
No response to the wake word	Wait until startup finishes (see the “Booting” section). Check the microphone array's USB cable.
It hears the wake word but gets my words wrong	Speak toward the front of the microphone from a few feet away, at normal volume. Wait for the light before speaking. Make sure you’re running in a quiet environment and as far away from the air compressor noise as possible.
It doesn't respond during a show	Expected: the character stops listening during shows. Stop the show from the web interface before you try speaking to it.
No sound at all	Check the speaker or amplifier power and the 3.5 mm cable. Headphones are not supported. If all else fails, restart the system.
Voice sounds robotic or different	The character is using its offline backup voice. Check the internet connection and the ElevenLabs key, voice ID and account credit to make sure your ElevenLabs subscription has not expired.
Answers are short and simple	The character is using the offline AI. Check the internet connection and the AI API key, and make sure your AI subscription has not expired.
It says it can't find or join WiFi	Check the network name and password in config.cfg, or join from the web interface.

Can't open the web interface	Make sure your phone or computer is on the same network. Ask the character for its IP address and type it in exactly, starting with http:// .
A show is missing from the list	Check the file names match exactly (see Adding custom shows) and the files are in the shows folder. Unplug and reinsert the drive.
Nothing moves	Check the air supply is on and the regulator reads the correct PSI.
Computer and/or MIDI sequencer can't recognize the character as an input/output source.	Make sure you're connecting your computer or MIDI sequencer to the character using a USB cable. The USB-C port on the character MUST be the one you use. Other USB-A ports on the character will not work for MIDI.
Program Blue running on a Windows PC doesn't recognize the character.	Ensure that you're using the proper cable adapter. For example, two USB to RS232 adapter cables, with a null modem adapter between the two cables. Connect the cables between the character and the Windows PC before starting Program Blue.
The animatronic character became sentient and wants to destroy all humans.	Hold your loved ones tight, and if you are religious, make peace with your god of choice.